



Tik-Tok limits use in China
The Chinese version of TikTok is limiting kids to 40 minutes a day
<https://digit.in/oct21-07>



Powerplant for mining
Bitcoin mining company buys Pennsylvania power plant to meet electricity needs.
<https://digit.in/oct21-09>

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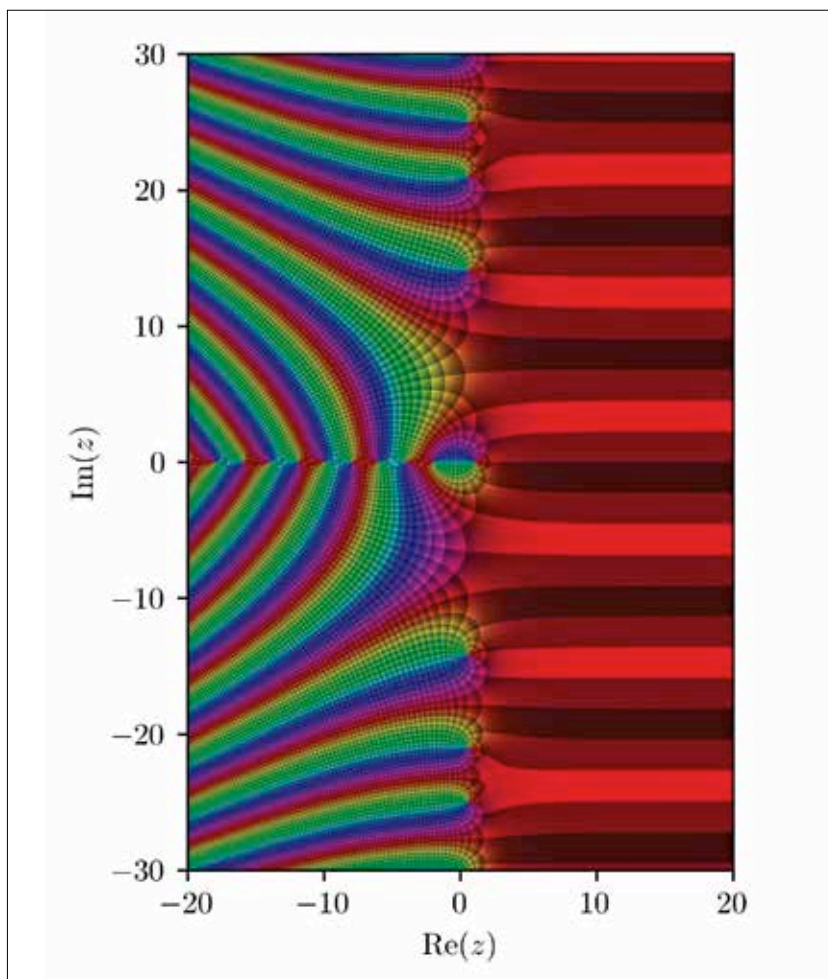
FEATURE

are represented on the x-axis or the horizontal line of the plane. As i is an imaginary number, they used the horizontal line or the so-called y-axis to represent the imaginary numbers, $-4i, -3i, -2i, -1i, 0, i, 2i, 3i, 4i...$ and so on. A complex number is simply made up of two parts, one is a real number and another is our imaginary number. We can represent every complex number as $a+ib$ where a and b are the real numbers and i is the imaginary unit which is equal to $\sqrt{-1}$, as we discussed earlier. Everything interesting in the Riemann hypothesis occurs between the line stretched at $\frac{1}{2}$ on the complex plane, between 0 and 1.

Let's come to functions. Think of it as a portal to another universe. You throw in a number and the portal spits out another different number. The letter f usually represents this portal in mathematics. If you put a number, x , in the portal, we will denote the number which it will spit out as $f(x)$. For example, let's see a cube function where every x we put in the portal, it'll spit out the cube of x . $f(x)=x^3$ therefore, $f(3) = 9$, $f(4) = 64$, $f(5) = 125$ and so on. This is a very simple function and is called a real function. This means if you throw in a real number, the portal will always throw out a real number. Now the mathematicians, not thinking straight as always, thought to put up complex numbers in place of x , for example, $f(i) = -1$. This was the beginning of what is called "complex functions". Here you can put in any complex number in the form of $a+ib$ and spit out any complex number.

Now, things will get interesting. There's such a similar complex function called the "Riemann Zeta function". This function is denoted by a Greek letter zeta, which is written as " ζ ". In this function, if you enter a complex number say, $a+ib$, you will get another complex number $c+ib$ as the result. Here's the Reimann Zeta Function:

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \frac{1}{1^s} + \frac{1}{2^s} + \frac{1}{3^s} + \dots$$



The Reimann Zeta function

Okay, so here comes the most interesting part. As we all know, a zero of the function is (obviously) a point $a+ib$ where $f(a+ib) = 0$. Like for $f(x) = x^3$, the only zero is at 0, where $f(0) = 0$. For Riemann Zeta Function, this is much more complicated. There are two zeroes in the Riemann zeta function, "Trivial" zeroes that occur at all the negative even integers such as $-2, -4, -6...$ and the other is "Non-Trivial" zeroes, which are the interesting zeroes. The ONLY thing we know about these non-trivial zeroes is that they occur at $\frac{1}{2} + ib$ for some b . All the non-trivial zeroes are found in the area called critical strip, between 0 and 1, precisely at $\frac{1}{2}$, which is known as the critical line. In the image above, the critical line is from 0 to -20, on which you can see a number of trivial zeroes.

Even after a lot of looking and testing, no other non-trivial zero is found outside the critical line. The Riemann hypothesis says that there are no non-trivial zeroes outside this area, but there is no proof that they are all like this. So, why is the Riemann Hypothesis so famous? The reason is prime number theory or prime number distribution is used in almost all the algorithms and encryption methods of the modern world in one or the other ways. Riemann's zeta function has been used by mathematicians to prove hundreds of theorems. New branches of number theory can open up with the proof of the Riemann hypothesis. New technologies, better security on the internet, quantum world, everything can be explored with the help of the Riemann hypothesis! [\[1\]](#)